Samarium

Sm (Sm I). Strongest classified transitions below 2 µm (NIST): 67 levels, 90 lines.

Sm^+ ionisation limit 45519.69 cm^{-1} = 5.64372 eV (219.685 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)
300
400
500
600
700
800
1000
1200
1500
2000
Energy above the ground state (cm^{-1}) — non-uniform scale
33 000
32 000
31 000
30 500
30 000
29 000
28 000
27 000
26 500
26 000
25 000
24 500
24 000
23 500
23 000
22 500
22 000
21 000
20 000
19 000
18 000
17 000
16 000
15 000
14 000
13 000
12 000
11 000
10 000
9 000
8 000
7 000
6 000
5 000
4 000
3 000
2 000
1 000
0
even
odd
$J = 0$
$J = 1$
$J = 2$
$J = 3$
$J = 4$
$J = 5$
$J = 6$
$J = 7$
$J = 8$
$^7F_0$ 0.000 cm^{-1} stable
$^7F_1$ 292.55
$^7F_2$ 811.872
$^7F_3$ 1489.50
$^7F_4$ 2273.6
$^7F_5$ 3125.41
$^7F_6$ 4020.62
^9H 11877.360 cm^{-1}
^9H 13...670 cm^{-1}
$^9H_7$
$^9H_8$ 14591.868
$^7D_1^o$ 26962.184 cm^{-1} τ = 24.8 ns
$(26804)_1^o$ 26803.634 cm^{-1} τ = 67.5 ns
$^7F_1^o$ 26281.063 cm^{-1} τ = 11.5 ns
$^7F_2^o$ 26786.751 cm^{-1} τ = 12.7 ns
$(23547)_2^o$ 23546.469 cm^{-1} τ = 31.1 ns
$^7G_2^o$ 23380.669 cm^{-1} τ = 11 ns
$^7G_1^o$ 22914.043 cm^{-1} τ = 7.3 ns
$(22893)_2^o$ 22893.345 cm^{-1} τ = 27.7 ns
$^9I_2^o$ 22843.943 cm^{-1} τ = 27.5 ns
$^5G_2^o$ 22491.899 cm^{-1} τ = 31.4 ns
$^7F_1^o$ 22313.629 cm^{-1} τ = 11.1 ns
$^7F_0^o$ 22041.017 cm^{-1} τ = 10.4 ns
$^7D_2^o$ 21813.211 cm^{-1} τ = 7.7 ns
$^7D_1^o$ 21193.640 cm^{-1} τ = 7.5 ns
$^5F_2^o$ 20762.804 cm^{-1} τ = 43.4 ns
$^5F_1^o$ 20090.935 cm^{-1} τ = 47 ns
$^7D_3^o$ 28356.189 cm^{-1} τ = 29.5 ns
$^7F_3^o$ 27425.328 cm^{-1} τ = 10.1 ns
$(24467)_3^o$ 24467.267 cm^{-1} τ = 63.4 ns
$^9H_3^o$ 24244.738 cm^{-1} τ = 38.2 ns
$^7H_3^o$ 24150.732 cm^{-1} τ = 21.4 ns
$(23710)_3^o$ 23709.922 cm^{-1} τ = 15.6 ns
$(23317)_3^o$ 23316.546 cm^{-1} τ = 22.5 ns
$^9I_3^o$ 23230.674 cm^{-1} τ = 36.2 ns
$^7D_3^o$ 22632.269 cm^{-1} τ = 8.2 ns
$^5F_3^o$ 21700.876 cm^{-1} τ = 34.3 ns
$^7G_3^o$ 19776.894 cm^{-1} τ = 38.7 ns
$^7D_4^o$ 29364.983 cm^{-1} τ = 32 ns
$^7F_4^o$ 28180.913 cm^{-1} τ = 10.3 ns
$(25387)_4^o$ 25387.153 cm^{-1} τ = 68.1 ns
$(24634)_4^o$ 24633.707 cm^{-1} τ = 10 ns
$(24312)_4^o$ 24311.948 cm^{-1} τ = 41.3 ns
$(23997)_4^o$ 23996.560 cm^{-1} τ = 22.9 ns
$^7D_4^o$ 23594.803 cm^{-1} τ = 8.7 ns
$^5F_4^o$ 22892.940 cm^{-1} τ = 24.1 ns
$(20713)_4^o$ 20712.769 cm^{-1} τ = 33.3 ns
$^7D_5^o$ 30233.580 cm^{-1} τ = 25.6 ns
$(29070)_5^o$ 29069.865 cm^{-1} τ = 18.5 ns
$(25616)_5^o$ 25615.537 cm^{-1} τ = 11.5 ns
$(25391)_5^o$ 25390.821 cm^{-1} τ = 60.1 ns
$(24669)_5^o$ 24668.734 cm^{-1} τ = 10 ns
$^7D_5^o$ 24323.421 cm^{-1} τ = 12.7 ns
$^5F_5^o$ 21809.669 cm^{-1} τ = 33.7 ns
$^7F_6^o$ 30092.414 cm^{-1} τ = 15.3 ns
$(29803)_6^o$ 29803.346 cm^{-1} τ = 34.4 ns
$(26471)_6^o$ 26471.335 cm^{-1} τ = 16 ns
$(26181)_6^o$ 26180.964 cm^{-1} τ = 16.5 ns
$(25840)_6^o$ 25840.296 cm^{-1} τ = 13.3 ns
$^7F_6^o$ 22944.337 cm^{-1} τ = 27.2 ns
$(22643)_6^o$ 22643.100 cm^{-1} τ = 38 ns
$(33507)_7^o$ 33507.230 cm^{-1} τ = 20.5 ns
$^5H_7^o$ 30330.790 cm^{-1} τ = 51.8 ns
$^7G_7^o$ 27287.496 cm^{-1} τ = 8.6 ns
$(23734)_7^o$ 23734.345 cm^{-1} τ = 31.5 ns
$^7H_7^o$ 23337.354 cm^{-1} τ = 47.5 ns
$^9I_8^o$ 31129.740 cm^{-1} τ = 32.9 ns
374.958 nm | 1.268 ea_0
380.502 nm | 1.484 ea_0
384.737 nm | 1.051 ea_0
392.631 nm | 2.234 ea_0
377.441 nm | 1.709 ea_0
395.300 nm | 2.642 ea_0
383.389 nm | 1.603 ea_0
372.208 nm | 2.183 ea_0
385.567 nm | 1.752 ea_0
397.578 nm | 3.421 ea_0
1.516 ea_0
369.113 nm | 1.711 ea_0
374.652 nm | 2.328 ea_0
385.983 nm | 2.717 ea_0
381.104 nm | 1.849 ea_0
399.114 nm | 3.489 ea_0
381.491 nm | 2.455 ea_0
581.649 nm | 4.095 ea_0
385.439 nm | 2.424 ea_0
383.556 nm | 4.507 ea_0
387.857 nm | 3.004 ea_0
507.810 nm | 5.219 ea_0
528.664 nm | 3.843 ea_0
580.211 nm | 5.002 ea_0
577.537 nm | 6.569 ea_0
422.736 nm | 1.835 ea_0
418.451 nm | 2.025 ea_0
439.459 nm | 1.997 ea_0
441.282 nm | 2.461 ea_0
450.037 nm | 3.272 ea_0
466.486 nm | 3.122 ea_0
438.165 nm | 2.593 ea_0
444.640 nm | 5.044 ea_0
444.305 nm | 3.525 ea_0
433.735 nm | 4.918 ea_0
432.567 nm | 2.489 ea_0
484.306 nm | 7.734 ea_0
429.796 nm | 8.257 ea_0
458.302 nm | 5.673 ea_0
488.534 nm | 6.232 ea_0
436.414 nm | 2.530 ea_0
439.858 nm | 1.917 ea_0
443.090 nm | 2.938 ea_0
432.074 nm | 3.150 ea_0
447.214 nm | 4.734 ea_0
428.403 nm | 3.614 ea_0
464.937 nm | 2.647 ea_0
444.352 nm | 2.913 ea_0
433.123 nm | 2.885 ea_0
428.470 nm | 2.383 ea_0
442.058 nm | 2.883 ea_0
452.444 nm | 1.644 ea_0
459.803 nm | 2.146 ea_0
448.157 nm | 1.939 ea_0
465.078 nm | 3.014 ea_0
467.206 nm | 2.503 ea_0
468.287 nm | 2.435 ea_0
453.507 nm | 2.899 ea_0
450.464 nm | 2.119 ea_0
428.340 nm | 4.513 ea_0
475.206 nm | 2.294 ea_0
446.054 nm | 2.276 ea_0
471.743 nm | 5.214 ea_0
451.257 nm | 2.893 ea_0
458.287 nm | 2.399 ea_0
467.215 nm | 2.887 ea_0
484.968 nm | 3.354 ea_0
464.670 nm | 2.647 ea_0
476.160 nm | 4.064 ea_0
492.036 nm | 3.403 ea_0
471.840 nm | 2.482 ea_0
478.444 nm | 3.296 ea_0
490.635 nm | 2.173 ea_0
472.975 nm | 4.339 ea_0
491.178 nm | 4.663 ea_0
488.514 nm | 2.481 ea_0
478.721 nm | 2.794 ea_0
469.004 nm | 3.883 ea_0
440.240 nm | 3.556 ea_0
497.737 nm | 1.818 ea_0
492.543 nm | 3.067 ea_0
445.420 nm | 3.532 ea_0
527.287 nm | 3.044 ea_0
520.203 nm | 4.207 ea_0
511.859 nm | 4.483 ea_0
512.356 nm | 3.941 ea_0
504.568 nm | 4.111 ea_0
528.437 nm | 3.928 ea_0
517.686 nm | 4.655 ea_0
507.261 nm | 5.535 ea_0

Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$|\langle J\|er\|J'\rangle|$ (ea_0)	A (s^{-1})	Γ/2π up. (MHz)	branch (%)	Use
$^7F_6$ – $^7F_6^o$	383.5563	383.4475	781.6127	4.507	5.610e+07	10.4	85.8	
$^7F_2$ – $^7F_1^o$	392.6312	392.5201	763.5471	2.234	5.570e+07	13.84	64.1	
$^7F_4$ – $^7F_3^o$	397.5781	397.4657	754.0467	3.421	5.390e+07	15.76	54.4	
$^7F_6$ – $^7G_7^o$	429.7956	429.6747	697.5234	8.257	1.160e+08	18.51	99.8	
$^7F_0$ – $^7G_1^o$	436.4136	436.2910	686.9457	2.530	5.200e+07	21.8	38	
$^7F_1$ – $^7G_1^o$	442.0575	441.9334	678.1752	2.883	6.500e+07	21.8	47.4	
$^7F_5$ – $(25616)_5^o$	444.6397	444.5149	674.2369	5.044	5.330e+07	13.84	61.3	
$^7F_4$ – $(24634)_4^o$	447.2138	447.0883	670.3560	4.734	5.640e+07	15.92	56.4	
$^7F_6$ – $(25840)_6^o$	458.3020	458.1736	654.1374	5.673	5.210e+07	11.97	69.3	
$^7F_1$ – $^7F_0^o$	459.8026	459.6738	652.0025	2.146	9.600e+07	15.3	99.8	
$^7F_2$ – $^7F_1^o$	465.0783	464.9481	644.6065	3.014	6.100e+07	14.34	67.7	
$^7F_2$ – $^7D_2^o$	476.1601	476.0270	629.6043	4.064	6.200e+07	20.67	47.7	
$^7F_1$ – $^7D_1^o$	478.4440	478.3103	626.5988	3.296	6.700e+07	21.22	50.2	
$^7F_6$ – $(24669)_5^o$	484.3058	484.1705	619.0149	7.734	9.700e+07	15.92	97	
$^7F_5$ – $^7D_4^o$	488.5344	488.3980	613.6568	6.232	7.500e+07	18.29	65.2	
$^7F_4$ – $^7D_3^o$	491.1777	491.0406	610.3544	4.663	5.310e+07	19.41	43.5	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{147}Sm hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$^7F_1$	-33.493	-58.688	F=5/2: +119.28 F=7/2: +75.41 F=9/2: -131.90
$^7G_1^o$	-212.620	-9.630	F=5/2: +951.63 F=7/2: +219.50 F=9/2: -746.58

Measured A, B constants; sources in the data file.